

Restrained critical apparatus

Pierre Deligne

Les intersections complètes de niveau de Hodge un

D013 / P03 / complete through physical EOF

This apparatus is editorial and is not part of Deligne’s article. The sole controlling source is the attached fourteen-page French authority, SHA-256 9A9C5C20DADB7D812FC3EF66D3EB7D254DEB27EEDCB13697B3E1EEB48243A3ED. P03 consumes the accepted P02 state and admits physical pages 12–14, printed 248–250, collected 300–302. The cumulative readers contain the complete article, ending immediately after “(Reçu le 26 août 1971)” / “(Received 26 August 1971)” at the authority’s physical EOF.

Accepted inherited corrections

source page/address	inherited error or omission	source-backed correction/disposition
physical 1 / printed 237 / collected 289	Upper-case title; strict > 3 ; “aux points”; H_n in the Jacobian lattice; prose substituted for “(=principalement polarisée)”; modernized source parentheses and literal “i.e.”.	Restore the exact title capitalization, $n = 2m + 1$ impair ≥ 3 , “en les points”, H^n throughout the quotient, “(=principalement polarisée)”, the source parentheses, and literal “i.e. indépendante”; carry the mathematical corrections into English.
physical 2 / printed 238 / collected 290	Parenthetical qualifiers dropped; the transition after (1.1.3) omitted; the Tate inverse system simplified.	Restore the qualifiers, the transition sentence, $\mathbb{Z}(1)_{\mathbb{Z}/r} =_{\text{dfn}} \mathbb{Z}(1)_{\mathbb{Z}} \otimes \mathbb{Z}/r$, the inverse limit indexed by r , and transition maps $x \mapsto x^{\ell^{r'-r}}$.
physical 3 / printed 239 / collected 291	$H^k(X, \mathbb{C})$ substituted for $H^k(X, \Omega_X^*)$; French inserted “a” after (1.3.2); “(unique)”, α , and citation forms normalized.	Restore Ω_X^* , preserve the printed French “Il un sens purement algébrique”, restore “une (unique)”, source α , and the visible citations; translate the French sentence grammatically in English.
physical 4 / printed 240 / collected 292	Visible $\mathbb{Z}_\ell(1)$ twists in (1.6.2) replaced by mathematically expected $\mathbb{Z}_\ell(m)$; left-subscript polarization notation, Ω^* , literal “connection(s)”, and references regularized.	Preserve the authority’s $\mathbb{Z}_\ell(1)$ factors, ${}_s\psi_\ell$, ${}_s\psi_{\text{DR}}$, Ω^* , literal French “connection(s)”, and the printed references without plausibility substitution.
physical 5 / printed 241 / collected 293	ι replaced by t ; “(très)” and [9] p. 19 / EGA III omitted; the triangle flattened; the H^0 map paraphrased and p_* changed to π_* .	Restore ι , “(très) ample”, the citation, the $Y \hookrightarrow \mathbb{P}_T^{n+v}$ triangle with j, g, p , the source H^0 map, and $p_*\mathcal{O}(k) \rightarrow g_*\mathcal{O}(k)$.
physical 6 / printed 242 / collected 294	The visible pair $\overline{S}'_{\mathbb{Z}}, S'_{\mathbb{Z}}$ was recast as $S'_{\mathbb{Z}}, S_{\mathbb{Z}}^0$; “resp.” and the Lefschetz citation expanded; the inherited reader overran the assigned page.	Restore the visible scheme symbols and roles, the source wording and “[6] Ch V, III p. 106”, and terminate the P01 readers exactly after “L’isomorphie de deux” / “The isomorphism of two”.
physical 7 / printed 243 / collected 295	The lattice intersection $v''^{-1}(H^1(A_s, \mathbb{Z})) \cap H^1(J(X_s), \mathbb{Z})$ was omitted; s/S and literal “i.e.” were normalized; ${}_s\psi$, coefficient fields, and the theorem triangle were altered, including $\mathbb{P}_V^r \mapsto \mathbb{P}_{\text{Spec}(V)}^N$.	Restore the full lattice expression, the visible s/S distinction and “i.e.”, all left-subscript polarization forms and scalar fields, “dédire de là”, and the exact $X \hookrightarrow \mathbb{P}_V^r \rightarrow \text{Spec}(V)$ triangle.
physical 8 / printed 244 / collected 296	The theorem’s split condition was compressed; “ S_K (1.8)” was shortened; ${}_s\psi$ was flattened; the repeated a in 2.3(a), guillemets, and the Bourbaki citation were regularized.	Restore the two-part (*) display and theorem wording, the 1.8 cross-reference, ${}_s\psi$ in 2.2, “un nombre a , a ayant”, the source quotation punctuation, and “Bourbaki, Alg Chap 2 §8 Th 1”.
physical 9 / printed 245 / collected 297	The printed French anomalies “Pout tout entier” and “ ℓ -adique 2.5 de” were silently corrected; $S_{\mathbb{Z}[1/b]}$ was normalized to $S_{\mathbb{Z}}[1/b]$; the compatibility parenthesis was absorbed; $M_{\ell_0^\infty} = \varprojlim_{\mathbb{C}} M_{\ell_0^\infty}$ became a generic M_∞ ; geometric fibers $F_{\overline{K}}$ became $F_{\overline{\eta}}$; the prime on J'_s was dropped; the point-count display was regularized by adding an s to q and $n + v$ to $\mathbb{P}$.	Preserve “Pout tout entier” and “ ℓ -adique 2.5 de” diplomatically; restore the source base-change notation, parenthetical sentence, exact ℓ_0 -power inverse system, every $\overline{K}$ fiber/fundamental-group subscript, the source J'_s , and exactly $q^{-m}(\#\mathbb{P}(k(s)) - \#X_s(k(s)))$, with no added subscript on q or superscript on $\mathbb{P}$.

physical 10 / printed 246 / collected 298	The two displayed trace identities at the page opening were reduced to prose; “Čebotarev” was modernized; the printed $\mathrm{GL}(R^n f_* \mathbb{Z}_\ell(m))$ in the proof of 2.6 was silently changed to the expected mod- ℓ term; b became b_ℓ .	Restore both Frobenius-trace displays, source spelling, the printed $\mathbb{Z}_\ell(m)$ proof term despite the lemma’s $\mathbb{Z}/\ell(m)$, and the literal scalar $b \in \mathbb{Z}_\ell^*$.
physical 11 / printed 247 / collected 299	French “repecte” was corrected; ${}_s\psi_{\mathrm{DR}}$ and (1.7.1) citation form were flattened; “(2.7.b)” was normalized; problem (I, K) and $\mu : G_K \times_K S_K \rightarrow S_K$ became a β -display without $\times_K$; the inherited reader continued into page 12.	Preserve the source typo diplomatically, all left-subscript forms and visible citation syntax, restore (I, K) , μ , $\times_K$, and μ^*/pr_2^* , and terminate P02 exactly after item 2.12(b) at the physical page-11 boundary.
physical 12 / printed 248 / collected 300	The continuation 2.12(c)–(d) lost the nested (α_ℓ) parentheses; every tilde on $\tilde{S}, \tilde{J}, \tilde{S}_k$ was stripped; $S_{\mathbb{Z}[1/k]}$ and “2.3.e)” were normalized; the visible $\tilde{S} = S$ in 3.3 was replaced by $S = S_\mathbb{Z}$.	Restore 2.12(c)–(d), $(J, (\alpha_\ell))$, every tilded scheme, the source base and cross-reference notation, $\tilde{S}_{k\mathbb{Q}}$, and the printed conjectural equality $\tilde{S} = S$.
physical 13 / printed 249 / collected 301	Tildes in 3.4–3.5 were removed; the extension point x'' and the bars on $\bar{X}, \bar{J}$ were omitted; “(géométriques)” was recast; bibliography entries 1–8 were silently regularized, including “Period” to “Periods” and source language/punctuation.	Restore $a : \tilde{J} \rightarrow \tilde{S}, \bar{x}, x'', \bar{X}, \bar{J}$, the parenthetical Frobenius qualifier and (3.5.1), plus the visible bibliography wording, capitalization, languages, the $P.A.$, : punctuation and separated double stop in item 1, the same $P.A.$, : in item 5, singular “Period”, and “Chelsea publ. Co.”.
physical 14 / printed 250 / collected 302	Entry 10 capitalization and quotation were modernized; EGA/SGA punctuation and the parenthesized distribution note were changed; the institution name and received matter were normalized or at risk of truncation; physical EOF was not proved.	Restore entries 9–11, lower-case “la” in item 10, “EGA. Eléments...”, “SGA 7 bis ... (Diffusé par l’IHES.)”, the authority’s address lines, “(Reçu le 26 août 1971)”, and terminate at the physical EOF.

Pagewise replay evidence for P03

- Page 12.** Completion of 2.12(c)–(d); section 3 opening; Proposition 3.1; Lemma 3.2 and proof; construction and gluing of $\tilde{S}_k$; conjecture 3.3. Every tilde and base-change subscript was checked against the rendered page.
- Page 13.** Proposition 3.4; complete 3.5, Witt-vector lift, x'' , $\bar{X}/\bar{J}$, geometric Frobenius, and (3.5.1); bibliography entries 1–8 with source wording and punctuation.
- Page 14.** Bibliography entries 9–11; exact EGA and SGA 7 bis notes; Pierre Deligne’s IHES address; received line; physical EOF.

Gate reconciliation

All frontmatter, units 1.1–1.13, units 2.1–2.12, units 3.1–3.5, list, backmatter, labelled-display, triangle, action/embedding, unlabelled-mathematics, and citation gates are accepted. The P02 partial 2.12 gate is closed by page 12. Seeded backmatter defect D013-005 is resolved by page-14 replay and EOF verification.

Publication-copy disposition

Journal line, copyright line, running headers, printed folios, and the page-1 printer mark are publication furniture and remain excluded from reader bodies. No authorial bibliography, note, address, date, or terminal matter is excluded.

Uncertainty and image fallback

No P03 locus required image fallback. All glyphs, formulae, tildes, bars, bibliography characters, and terminal matter were safely encoded. The uncertainty/image ledger remains header-only.

Final state disposition

P03 is complete through physical EOF. Accepted physical pages: 1–14. First untouched physical page: none. The full cold audit is closed after the page-9 source-backed correction.